

FRD

Functional Requirements Document (FRD)

AWS S3-Based Data Processing Pipeline

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FR-INGEST-001

Title

Ingest Customer and Order CSV Files from S3

Description

The system shall read CSV files from designated S3 buckets for customer and order data.

Preconditions

- S3 buckets `s3://adif-sdlc/sdlc\_wizard/customerdata/` and `s3://adif-sdlc/sdlc\_wizard/orderdata/` exist and are accessible
- CSV files conform to defined schemas
- AWS Glue has appropriate IAM permissions to read from source buckets

Main Flow / Functional Steps

- Read customer CSV files from `s3://adif-sdlc/sdlc\_wizard/customerdata/`
- Parse customer data with schema: CustId, Name, EmailId, Region
- Read order CSV files from `s3://adif-sdlc/sdlc\_wizard/orderdata/`
- Parse order data with schema: OrderId, ItemName, PricePerUnit, Qty, Date, CustId

Postconditions

- Customer and order datasets loaded into memory/dataframes for processing
- Data ready for cleaning operations

Assumptions

- CSV files are well-formed and delimited consistently
- File encoding is UTF-8 or compatible
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FR-CLEAN-001

Title

Remove Null Values from Customer and Order Datasets

Description

The system shall remove all records containing "Null" string values or NULL values from both customer and order datasets.

Preconditions

- Customer and order datasets loaded from S3

Main Flow / Functional Steps

- Scan customer dataset for "Null" string and NULL values across all columns
- Remove rows containing "Null" or NULL in any column
- Scan order dataset for "Null" string and NULL values across all columns
- Remove rows containing "Null" or NULL in any column

Postconditions

- Customer and order datasets contain no "Null" or NULL values
- Cleaned datasets ready for duplicate removal

Assumptions

- "Null" refers to the literal string "Null" (case-sensitive or case-insensitive as per implementation)
- NULL refers to standard null/empty values
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FR-CLEAN-002

Title

Remove Duplicate Records from Customer and Order Datasets

Description

The system shall remove duplicate records from customer and order datasets independently.

Preconditions

- Null values removed from both datasets

Main Flow / Functional Steps

- Identify duplicate rows in customer dataset based on all columns
- Retain first occurrence, remove subsequent duplicates
- Identify duplicate rows in order dataset based on all columns
- Retain first occurrence, remove subsequent duplicates

Postconditions

- Customer dataset contains unique records only

- Order dataset contains unique records only
- Cleaned datasets ready for catalog registration

Assumptions

- Duplicates are identified by exact match across all columns
- No specific business key prioritization for duplicate resolution
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FR-CATALOG-001

Title

Register Customer and Order Tables in Glue Data Catalog

Description

The system shall register cleaned customer and order datasets as tables in AWS Glue Data Catalog.

Preconditions

- Customer and order datasets cleaned (nulls and duplicates removed)
- Glue database `gen\_ai\_poc\_databrickscoe` exists

Main Flow / Functional Steps

- Create or update table `sdhc\_wizard\_customer` in database `gen\_ai\_poc\_databrickscoe`
- Register customer schema: CustId, Name, EmailId, Region
- Create or update table `sdhc\_wizard\_order` in database `gen\_ai\_poc\_databrickscoe`
- Register order schema: OrderId, ItemName, PricePerUnit, Qty, Date, CustId

Postconditions

- Tables `sdhc\_wizard\_customer` and `sdhc\_wizard\_order` queryable via Athena
- Metadata available in Glue Data Catalog

Assumptions

- Glue database `gen\_ai\_poc\_databrickscoe` pre-exists or is created as part of setup
- Table schemas match source CSV schemas exactly
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FR-SCD-001

Title

Implement SCD Type 2 for Order Summary Using Hudi

Description

The system shall maintain a Slowly Changing Dimension Type 2 table for order summary data using Apache Hudi format on S3.

Preconditions

- Customer and order tables registered in Glue catalog
- S3 location `s3://adif-sdlc/curated/sdlc\_wizard/ordersummary/` accessible
- Hudi libraries available in Glue environment

Main Flow / Functional Steps

- Join customer and order datasets on CustId
- Detect changes in joined records compared to existing ordersummary table
- For new records: insert with IsActive=true, StartDate=current timestamp, EndDate=null, OpTs=current timestamp
- For changed records: close previous row (set IsActive=false, EndDate=current timestamp), insert new active row
- Write results to `s3://adif-sdlc/curated/sdlc\_wizard/ordersummary/` in Hudi format

Postconditions

- ordersummary table contains current and historical versions of records
- Active records marked with IsActive=true
- Historical records marked with IsActive=false and populated EndDate
- Table registered as `ordersummary` in Glue catalog database `gen\_ai\_poc\_databrickscoe`

Assumptions

- Change detection based on comparison of all business columns
- OpTs represents operation timestamp
- StartDate and EndDate represent validity period of each record version
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FR-SCD-002

Title

Trigger Incremental SCD2 Updates on Customer Changes

Description

The system shall detect customer data updates and trigger corresponding SCD Type 2 updates in the ordersummary table.

Preconditions

- ordersummary table exists with SCD2 structure
- Customer updates available in source location

Main Flow / Functional Steps

- Identify changed or new customer records since last processing
- Retrieve affected orders linked to changed customers via CustId
- Apply SCD2 logic to ordersummary for affected records
- Maintain IsActive, StartDate, EndDate, OpTs columns during update

Postconditions

- ordersummary reflects latest customer information
- Historical versions preserved with proper temporal attributes
- Incremental processing completed without full table rewrite

Assumptions

- Customer changes detected via timestamp or change data capture mechanism
- Only affected records in ordersummary are updated
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FR-AGG-001

Title

Generate Customer Aggregate Spend Table

Description

The system shall create and maintain an aggregation table showing total spending per customer.

Preconditions

- ordersummary table populated with active records
- S3 location `s3://adif-sdlc/analytics/customeraggregatespend/` accessible

Main Flow / Functional Steps

- Query active records from ordersummary (IsActive=true)
- Calculate TotalAmount per customer: sum(PricePerUnit Qty) grouped by customer Name
- Include Date field in aggregation
- Write results to `s3://adif-sdlc/analytics/customeraggregatespend/`
- Register table as `customeraggregatespend` in Glue catalog database `gen\_ai\_poc\_databrickscoe`

Postconditions

- customeraggregatespend table contains columns: Name, TotalAmount, Date
- Table queryable via Athena
- Aggregated data reflects current active order summary records

Assumptions

- TotalAmount calculated as PricePerUnit multiplied by Qty, summed per customer
- Date field represents aggregation date or latest order date
- Only active ordersummary records included in aggregation
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FR-ARCH-001

Title

Utilize AWS Architecture Components for Pipeline

Description

The system shall leverage specified AWS services for storage, processing, cataloging, querying, and monitoring.

Preconditions

- AWS account with appropriate service quotas
- IAM roles and policies configured

Main Flow / Functional Steps

- Use S3 for all data storage (source, curated, analytics layers)
- Use AWS Glue with Spark and Hudi for ETL processing
- Use Glue Data Catalog for metadata management
- Use Athena for ad-hoc querying of cataloged tables
- Use CloudWatch for pipeline monitoring and logging
- Optionally use Lake Formation for data governance

Postconditions

- Pipeline operational using designated AWS services
- Monitoring and logging active via CloudWatch
- Data governance enforced if Lake Formation enabled

Assumptions

- All services deployed in same AWS region for optimal performance
- Lake Formation is optional and not mandatory for initial implementation
- CloudWatch logs retained per organizational policy
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ASSUMPTIONS

- CSV delimiter is comma; no special handling for embedded delimiters required
- All timestamps use UTC timezone
- CustId is the primary key for customer data
- OrderId is the primary key for order data
- CustId in order data is a valid foreign key to customer data
- Hudi configuration (record key, precombine field, partition) will be defined during technical design
- Incremental processing frequency not specified; assumed to be batch-based
- No real-time streaming requirements
- Data retention policies not specified; all historical SCD2 records retained indefinitely
- No data masking or encryption requirements specified beyond standard S3 encryption
- Athena query performance optimization (partitioning, bucketing) deferred to technical design
- Error handling and retry logic deferred to technical design
- No specific data quality validation rules beyond null and duplicate removal